

Derivatives — group rounds

Lessons 1–4, 7–12 · Quarter I · 40 min · 20 questions · 100 points · groups of 3–4

Four rounds, each timed and worth more than the last. The tag on every question names the topic it comes from. Work on paper, hand in one sheet for the group. An answer with no method scores half.

GROUP

MEMBERS

SCORE

Round A · Easy5'2 pts each

Two from each topic — speed round

- 1

T1

For $y = x^2$, write the increment Δy as x goes from 3 to $3 + h$.
- 2

T1

Find the slope of the secant to $y = \frac{1}{x}$ between $x = 1$ and $x = 2$.
- 3

T2

Evaluate $\lim$ as $x \rightarrow 3$ of $\frac{x^2 - 9}{x^2 - 2x - 3}$.
- 4

T2

Evaluate $\lim$ as $x \rightarrow 0$ of $\frac{\sqrt{x+1} - 1}{x}$.
- 5

T3

Differentiate $y = \frac{2x^3 - 5}{x^2}$.
- 6

T3

Differentiate $y = x\sqrt{x}$.
- 7

T4

Differentiate $y = \frac{1}{(2x - 7)^3}$.
- 8

T4

Differentiate $y = \sqrt[3]{3x + 1}$.

Round B · Medium8'4 pts each

Rewrite before you differentiate

- 9

T1

Write the secant slope of $y = \sqrt{x}$ between $x = 4$ and $x = 4 + h$ without a surd in the numerator, then state its limit as $h \rightarrow 0$.
- 10

T2

Evaluate $\lim$ as $x \rightarrow 2$ of $\frac{x^3 - 8}{x - 2}$.
- 11

T2

Evaluate $\lim$ as $x \rightarrow 1$ of $\frac{1}{x - 1} - \frac{2}{x^2 - 1}$.
- 12

T3

Differentiate $y = (x + 1)(x - 2)(x + 3)$.
- 13

T3

Differentiate $y = \frac{x^2 + 1}{x^2 - 1}$.
- 14

T4

Differentiate $y = (1 - 2x)^6(x + 3)$ and factorise.

Round C · Hard9'7 pts each

Two steps at least — be ready to explain

- 15

T1 T2

For $f(x) = \frac{1}{x^2}$, write the secant slope between $x = 2$ and $x = 2 + h$ as a single fraction, simplify it, and find its limit as $h \rightarrow 0$.
- 16

T2

Evaluate $\lim$ as $x \rightarrow 4$ of $\frac{\sqrt{x} - 2}{x^2 - 16}$.
- 17

T3

The curve $y = \frac{x^2 - 1}{x^2 + 1}$ has a horizontal tangent. Find where, and the value of y there.
- 18

T4

Differentiate $y = \sqrt{\frac{2x + 1}{2x - 1}}$ for $x > \frac{1}{2}$.

Round D · Very hard8'16 pts each

Every part needed for full marks

- 19

T3 T4

(a) Show that $y = \frac{x}{\sqrt{x^2 + 4}}$ has $y' = \frac{4}{(x^2 + 4)^{3/2}}$. (b) Hence find the equation of the tangent at $x = 0$. (c) Explain why the curve has no stationary point.

20 T1 T3 A stone is thrown upwards and its height is $h(t) = 25t - 5t^2$ metres after t seconds.

(a) Write the average velocity between $t = 1$ and $t = 1 + k$ as an expression in k , simplified. (b) State its limit as $k \rightarrow 0$, and check it against $h'(1)$ found by the rules. (c) Find the greatest height the stone reaches.
